What are Escape Sequences?

Escape sequences are special characters used inside a C++ string to perform a specific action or display special characters.

An escape sequence usually starts with a **backslash (`\`)**.

Common Escape Sequences

Escape Sequence	Name	Purpose
`\n`	Newline	Moves the cursor to the next line
`\t`	Tab	Adds a horizontal tab space
`\\`	Backslash	Displays a backslash `\`
`\"`	Double Quote	Displays a double quotation mark `"`
`\'`	Single Quote	Displays a single quotation mark `'`
`\r`	Carriage Return	Moves the cursor to the beginning of the current line

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1. Newline – `\n`

What is `\n`?

The `\n` escape sequence is used to **move the cursor to the next line**.

It is similar to using `endl`.

Question

Write a C++ program that prints two lines of text using the newline escape sequence `\n`.

Example 1

```cpp
#include <iostream>
using namespace std;

int main() {

    cout << "Hello World!\n";
    cout << "Welcome to C++.";

    return 0;
}
```

Output

```
Hello World!
Welcome to C++.
```

Understand the Code

```
cout << "Hello World!\n";
```

The `\n` moves the cursor to the next line. Therefore, the second `cout` statement starts on a new line.

Remember

```
\n → New line
```

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2. Tab – `\t`

What is `\t`?

The `\t` escape sequence inserts a **horizontal tab space**.

It is useful for creating simple columns in output.

Question

Write a C++ program that displays a list of items and their prices using `\t` to separate the columns.

Example 2

```
#include <iostream>
using namespace std;

int main() {

    cout << "Item\tPrice\n";
    cout << "Apple\t100\n";
    cout << "Mango\t150\n";

    return 0;
}
```

Output

```
Item    Price
Apple   100
```

```
Mango   150
```

Understand the Code

```
cout << "Apple\t100\n";
```

- `\t` adds a tab space between `Apple` and `100`.
- `\n` moves the cursor to the next line.

Remember

```
\t → Tab space
```

---

3. Backslash – \\

What is \\?

The backslash character `\` has a special meaning in C++.

If we want to **display an actual backslash**, we use two backslashes:

```
\\
```

Question

Write a C++ program that displays a Windows file path containing backslashes.

Example 3

```
#include <iostream>
using namespace std;

int main() {

    cout << "File path: C:\\Users\\Student\\Documents";

    return 0;
}
```

Output

```
File path: C:\Users\Student\Documents
```

Understand the Code

In the program:

```
C:\\Users\\Student\\Documents
```

Each `\\` represents **one actual backslash** in the output.

Remember

```
\\ → Displays \
```

---

4. Double Quote – `\"`

What is `\"`?

A double quotation mark `"` normally marks the beginning or end of a string.

If we want to display a double quotation mark **inside a string**, we use:

```
\"
```

Question

Write a C++ program that prints a sentence containing double quotation marks.

Example 4

```cpp
#include <iostream>
using namespace std;

int main() {

    cout << "He said, \"C++ is fun!\"";

    return 0;
}
```

Output

```
He said, "C++ is fun!"
```

Understand the Code

```
\"C++ is fun!\"
```

The \" tells C++ to display a quotation mark instead of treating it as the end of the string.

Remember

```
\" → Displays "
```

---

5. Single Quote – \'

What is \'?

The \' escape sequence can be used to display a **single quotation mark**.

Question

Write a C++ program that prints a character enclosed in single quotation marks.

Example 5

```
#include <iostream>
using namespac
```

6. Carriage Return – \r

In C++, \r means carriage return. It moves the cursor to the beginning of the current line without moving to a new line.

Example 6

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello\rWorld";
    return 0;
}
```

Understand the Code

- Hello is printed on the screen.

- `\r` moves the cursor to the beginning of the same line.
- `World` is printed from the beginning, overwriting the existing characters where applicable.

The output is typically:

```
World
```

Example 7: Easy-to-Understand Demonstration

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "ABCDEF\rXYZ";
    return 0;
}
```

Output:

```
XYZDEF
```

Here, `\r` returns the cursor to the beginning. `XYZ` replaces the first three characters, while `DEF` remains.

Remember the Difference

Escape sequence	Meaning
\n	Moves to a new line
\r	Moves to the beginning of the current line
\t	Adds a horizontal tab

Tip: The visible result of `\r` can vary depending on the console or terminal. It is often used to update progress messages on the same line.